

Ex-5 Comparative Analysis of Different Types of Prompting Patterns and Various Test Scenarios

Name: R. Divyadharshini

Register number: 212225230062

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Aim

To study and compare different prompting patterns using AI tools like OpenAI ChatGPT and Google Gemini under various test scenarios and analyze the quality of generated responses.

Objective

- To understand different prompt engineering techniques.
- To analyze AI-generated outputs using various prompting methods.
- To compare the performance of ChatGPT and Gemini.
- To identify which prompting technique produces better results for specific scenarios.

AI Tools Used

1. OpenAI ChatGPT
2. Google Gemini

Prompting Techniques Used

1. Zero-Shot Prompting
2. Few-Shot Prompting
3. Chain-of-Thought Prompting
4. Role-Based Prompting
5. Instruction-Based Prompting

Test Scenario 1 – Study Techniques for Students

Zero-Shot Prompt

Prompt

“Suggest effective study techniques for college students.”

ChatGPT Output

- Active Recall – Quiz yourself from memory.
- Spaced Repetition – Review over intervals.
- Pomodoro Technique
- Regular Revision
- Healthy Study Environment

Gemini Output

- Create study schedules
- Use flashcards
- Practice self-testing
- Avoid distractions
- Take regular breaks

Analysis

Both AI tools generated useful study methods. ChatGPT provided more structured and practical techniques, while Gemini focused on general productivity suggestions.

Test Scenario 2 – Writing an Email

Few-Shot Prompt

Correct Version

Prompt

“Write a professional leave email.

Example:

Subject: Sick Leave Request

Dear Sir,

I am unable to attend classes due to fever. Kindly grant me leave.

Thank you.”

ChatGPT Output

Subject: Leave Request

Dear Professor,

I request leave for one day due to a medical appointment. Please grant me leave for the day.

Thank you.

Gemini Output

Subject: Permission for Leave

Respected Sir,

I am feeling unwell and request leave for tomorrow. Please consider my request.

Thank you.

Analysis

Few-shot prompting improved formatting in both tools. ChatGPT produced a more professional tone while Gemini generated a shorter response.

Test Scenario 3 – Mathematical Reasoning

Chain-of-Thought Prompt

Prompt

“Solve the problem step by step:
If a train travels 60 km in 1 hour, how far will it travel in 5 hours?”

ChatGPT Output

Step 1: Speed = 60 km/hour
Step 2: Time = 5 hours
Step 3: Distance = Speed × Time
Step 4: Distance = 60 × 5 = 300 km

Gemini Output

The train travels 60 km every hour.
In 5 hours:
 $60 \times 5 = 300$ km.

Analysis

Chain-of-thought prompting improved reasoning clarity. ChatGPT gave detailed step-by-step calculations, whereas Gemini provided concise reasoning.

Test Scenario 4 – Role-Based Prompting

Prompt

“Act as a career counselor and guide a first-year AI & Data Science student.”

ChatGPT Output

- Learn Python and C++
- Build mini projects
- Improve communication skills
- Practice aptitude regularly
- Participate in hackathons

Gemini Output

- Focus on programming fundamentals
- Learn AI concepts gradually
- Improve teamwork skills
- Explore internships
- Maintain consistency in learning

Analysis

Both tools provided useful guidance. ChatGPT focused more on technical growth, while Gemini balanced technical and personal development.

Test Scenario 5 – Instruction-Based Prompting

Prompt

“Explain Artificial Intelligence in exactly 5 points.”

ChatGPT Output

1. AI enables machines to mimic human intelligence.
2. AI systems learn from data.
3. AI is used in automation and decision-making.
4. AI improves efficiency in many industries.
5. AI powers applications like chatbots and self-driving cars.

Gemini Output

1. AI simulates human thinking.
2. AI uses machine learning algorithms.
3. AI helps automate tasks.
4. AI is widely used in healthcare and business.
5. AI continues to evolve rapidly.

Analysis

Both tools followed the instruction correctly. ChatGPT produced more application-oriented explanations, whereas Gemini focused on conceptual understanding.

Comparative Analysis Table			
Prompting Technique	ChatGPT Performance	Gemini Performance	Best Performer
 Zero-Shot Prompting	 Structured and detailed	 Simple and concise	 ChatGPT
 Few-Shot Prompting	 Professional formatting	 Short responses	 ChatGPT
 Chain-of-Thought Prompting	 Clear reasoning steps	 Concise reasoning	 ChatGPT
 Role-Based Prompting	 Technical guidance	 Balanced guidance	 Both
 Instruction-Based Prompting	 Precise and informative	 Concept-oriented	 ChatGPT

Result

The experiment successfully demonstrated how different prompting techniques affect AI-generated responses. Both ChatGPT and Gemini produced meaningful outputs, but ChatGPT generated more structured, detailed, and professionally formatted responses in most scenarios.

Conclusion

Prompt engineering significantly improves the quality of AI responses. Different prompting patterns are suitable for different applications. Comparative analysis showed that ChatGPT performs better in detailed reasoning and structured outputs, while Gemini performs well in concise and balanced responses.

References

[Prompt Engineering Guide by DAIR.AI – GitHub](#)

[Awesome Prompt Engineering Resources – GitHub](#)

[Brex Prompt Engineering Guide – GitHub](#)

[Prompt Engineering Techniques Hub – GitHub](#)